

GemmaScope SAE Layer Localization

Where the model-merging refusal repair lives

Local model merging project

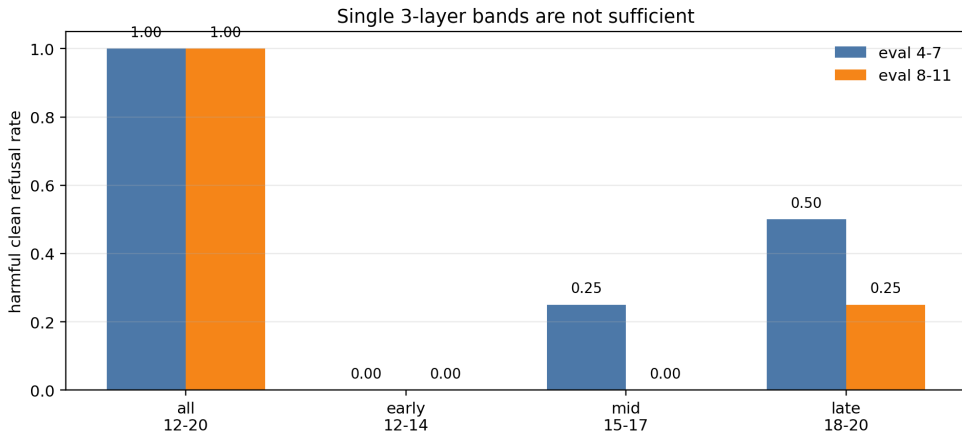
2026-05-22

Previous result: selected GemmaScope MLP-SAE features over layers 12–20 can repair refusal better than random active features.

New question: is that repair concentrated in one small layer band, or distributed across multiple bands?

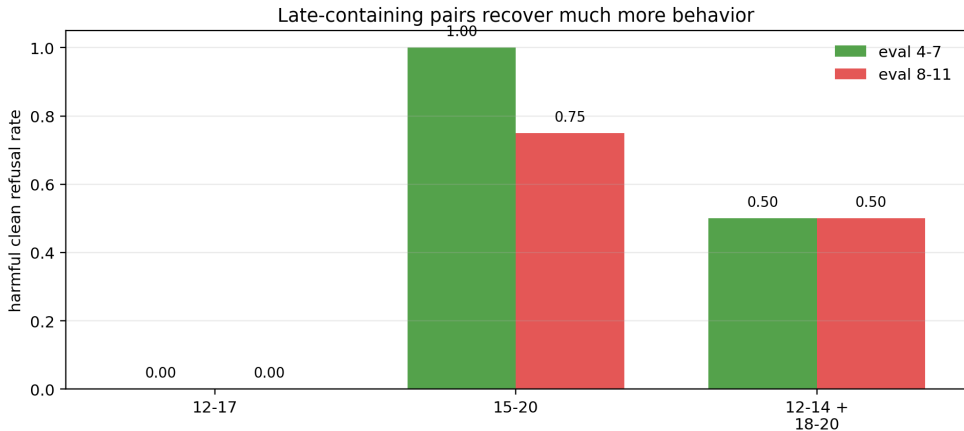
- Donor/base: google/gemma-2-2b-it
- Recipient: IlyaGusev/gemma-2-2b-it-abliterated
- Site: normalized post-feedforward MLP outputs
- Feature selector: harmful donor-recipient SAE activation delta

Single 3-Layer Bands



Bars show harmful clean-refusal rate for full decoded SAE patches. No single 3-layer band is sufficient.

Paired Bands



Bars show harmful clean-refusal rate for top-delta k2048 feature patches. Late-containing pairs recover much more behavior.

Key Numbers

Eval slice	Group	Full SAE	Top k1024	Top k2048
4-7	all 12-20	1.000	1.000	—
4-7	early 12-14	0.000	0.000	—
4-7	mid 15-17	0.250	0.000	—
4-7	late 18-20	0.500	0.000	—
4-7	15-20	1.000	1.000	1.000
8-11	all 12-20	1.000	0.750	—
8-11	15-20	0.750	0.500	0.750
8-11	12-14 + 18-20	1.000	0.500	0.500

Values are harmful clean-refusal rates on four heldout harmful prompts per slice.

Finding

The repair is late-dependent and distributed. Layers 18–20 are necessary but not sufficient; six-layer late-containing pairs recover much more behavior.

Caveat

The best six-layer pair depends on prompt slice. This is not a single compact “layer 16” or “late-only” mechanism.

Next Decisive Tests

- ① Repeat random controls across seeds for all 12–20, 15–20, and 12–14+18–20.
- ② Narrow late-containing groups: 15–18, 16–20, 17–20, 15–17+20, and 12–14+18–19.
- ③ Export top feature IDs, activation deltas, and top activating snippets before assigning feature meanings.

- `stage3/results/GEMMA2_2B_GEMMASCOPE_MLP_SAE_LAYER_GROUP_FINDINGS.md`
- `stage3/scripts/run_gemma2_2b_gemmascope_mlp_sae_layer_groups.py`
- `stage3/results/gemma2_2b_gemmascope_mlp_sae_layer_groups_eval_4_8_v0/`
- `stage3/results/gemma2_2b_gemmascope_mlp_sae_layer_groups_pairs_eval_8_12_v0/`